

0DTE 15:30–16:00 nearest-OTM package: per-rule return summary, models compared on the same 871 days

	n	mean	std	min	25%	50%	75%	max	skew	ex. kurt	t	Sharpe _{ann}	n_{buy}	%buy
<i>Short volatility (always short)</i>														
all models (no forecast)	871	0.020	1.127	-10.32	-0.444	0.329	0.931	1.00	-2.31	10.90	0.53	0.28	0	0.0
<i>Long-short volatility (± 1 on sign(s))</i>														
HAR + calendar OLS	871	-0.007	1.127	-10.32	-0.616	0.238	0.884	4.63	-1.91	10.68	-0.18	-0.10	144	16.5
block-diag ridge	871	0.004	1.127	-10.32	-0.588	0.264	0.896	4.63	-1.88	10.76	0.11	0.06	114	13.1
LightGBM	871	0.008	1.127	-10.32	-0.579	0.252	0.900	4.63	-1.90	10.78	0.20	0.11	131	15.0
XGBoost	871	-0.005	1.127	-10.32	-0.624	0.230	0.883	4.63	-1.85	10.70	-0.12	-0.07	140	16.1
lasso (causally tuned)	871	-0.011	1.127	-10.32	-0.623	0.229	0.884	4.63	-1.89	10.65	-0.29	-0.16	132	15.2
lasso ($\alpha = 10^{-4}$)	871	-0.002	1.127	-10.32	-0.616	0.234	0.884	4.63	-1.83	10.71	-0.05	-0.03	133	15.3
elastic net (causally tuned)	871	-0.013	1.127	-10.32	-0.605	0.230	0.884	4.63	-1.92	10.64	-0.35	-0.19	131	15.0
<i>Unit-median VRP size</i>														
HAR + calendar OLS	871	0.047	1.085	-6.84	-0.213	0.079	0.500	3.13	-1.21	5.68	1.28	0.69	144	16.5
block-diag ridge	871	0.059	1.049	-6.84	-0.220	0.085	0.484	3.16	-1.12	5.94	1.67	0.90	114	13.1
LightGBM	871	0.057	1.112	-6.84	-0.241	0.072	0.495	9.11	-0.54	9.85	1.52	0.82	131	15.0
XGBoost	871	0.052	1.095	-6.84	-0.235	0.080	0.502	5.84	-1.07	6.53	1.41	0.76	140	16.1
lasso (causally tuned)	871	0.060	1.088	-6.84	-0.210	0.085	0.513	3.00	-1.30	6.45	1.63	0.88	132	15.2
lasso ($\alpha = 10^{-4}$)	871	0.058	1.072	-6.84	-0.209	0.073	0.475	3.00	-1.27	6.53	1.59	0.86	133	15.3
elastic net (causally tuned)	871	0.048	1.077	-6.84	-0.221	0.080	0.493	3.00	-1.32	6.51	1.31	0.70	131	15.0

One expiration day = one return; mid fill. $t = \sqrt{n} \cdot \text{mean}/\text{std}$. “ex. kurt” is Fisher excess kurtosis (fourth standardized moment minus 3; Gaussian = 0). $n_{\text{buy}}/\% \text{buy}$ count days the rule is net long ($q_t > 0$). The short-volatility rule takes no forecast, so it is one row for all models. Sharpe_{ann} = (mean/std) $\times \sqrt{252}$; every other column is daily.

Forecasts: HAR + calendar OLS; block-diagonal ridge (HAR block and exogenous block, separate penalties); LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned).